



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	4YRS B.S	IIT Kharagpur	8.31 / 10
2023	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	91.2%
2021	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	97.6%

INTERNSHIPS AND PROJECTS

Analyst Programmer Intern | Xceedance Consulting Pvt. Ltd. (May'25 - Sept'25)
[Role Description: Extraction of Information from Insurance data using LLM and Normalization of insurance data]
• Acquired insurance data from unstructured sources and transformed it into datasets optimized for NER modeling.
• Normalized data by standardizing and parsing entity names, enforcing consistent schemas and formats across datasets.
• Data normalization verified each entity against a credible open-source dataset for use in bank documents.
• Built and deployed a **few-shot learning model** using the Azure AI API to accurately classify and extract key insurance entities.
• Achieved **94.2%** peak classification accuracy and a **0.91 weighted F1-score** across four insurance entity categories

Research Intern | Dept. Of Computer Science | (Under Prof. Pawan Goyal) (May'24-Sept'24)
[Role Description: Developed a Sanskrit ASR system by fine-tuning Wav2Vec2 on a custom-curated dataset.]
• Scraped a website using **BeautifulSoup** to gather a dataset of transcripts and corresponding audio files
• Cleaned and preprocessed the dataset to ensure consistency and accurate alignment between audio and text
• Trained a **Wav2Vec2** model on the prepared dataset to perform speech-to-text transcription for Sanskrit
• Achieved an average WER of 17.5% and CER of 6.2%, which is fairly high for complex linguistic asr system

Full-Stack Web Developer | Homecoming Cloud Kitchen (May '24-Oct'24)
[Role description: Designed and shipped a PWA for a cloud kitchen, owning frontend, Firebase backend, admin tooling]
• Built a **multi-page static PWA** with HTML/CSS/JS and a central app controller managing authorisation, routing, and UI state.
• Integrated **Firebase Authentication** and **Firestore**, delivering secure auth flows, protected routes, and real-time listeners.
• Developed a full Admin Dashboard with live orders, status workflows, menu CRUD, reports, and operator messaging.
• Shipped offline-ready PWA features using service workers, cache-first assets, manifest, dark mode, and shared theming.

PROJECTS

House Price Prediction | Kaggle Competition Project
• Engineered 200+ features via statistical imputation and **one-hot encoding**, increasing model performance by 30%.
• Developed and compared 6 regression models including **XGBoost**, achieving a 90% R-squared score and \$25,450 RMSE.
• Applied **Bayesian Optimization** to tune 9 hyperparameters, reducing prediction error by 50% using 5-fold CV.
• Conducted extensive EDA and feature importance analysis to secure a Kaggle rank of **2,111** in a global competition..

Real Time Emergency Evacuation System | Self Project
• Engineered a **YOLOv11** and HSV vision pipeline for real-time human and hazard detection at a consistent 30 FPS.
• Developed a spatial mapping system using **homography** to project 3D camera coordinates onto a 2D digital floor plan.
• Implemented the **A-Star pathfinding algorithm** to dynamically recalculate safe evacuation routes in under 50ms
• Optimized pipeline performance by leveraging **NumPy** for vectorized calculations and **OpenCV** for low-latency visualization.

Titanic Survival Prediction | Kaggle Competition Project
• Analyzed survival patterns for 891 samples, uncovering key correlations between gender and class to inform feature selection.
• Engineered a pipeline using statistical imputation and one-hot encoding for categorical variable alignment.
• Developed an **XGBoost classifier** achieving a 0.88 ROC-AUC by tuning 6 hyperparameters for optimal prediction.
• Secured a Kaggle rank of **3,916** with 0.775 accuracy using feature importance and confusion matrix evaluation protocols.

AWARDS AND ACHIEVEMENTS

- Ranked among the **top 6.4% in JEE Advanced 2023** and **top 1.9% in JEE Mains 2023**.
- Ranked among the **top 1% in WBJEE** (West Bengal Joint Entrance Exam) among more than 1 Lakh students.
- Achieved a **peak rating of 1400+ (Specialist) on Codeforces** (User Handle : **kingdhruv**).
- Solved **150+** Leetcode Problems consistently improving my coding abilities and logical reasoning

SKILLS AND EXPERTISE

C | C++ | Python | HTML | CSS | Pandas | NumPy | Matplotlib | scikit-learn | TensorFlow | BeautifulSoup4 | VS Code | spaCy | Canva | Sublime Text | Git | Github | Large Language Models | MySQL | Javascript | Bash | Jupyter Notebook | MS Office

COURSEWORK INFORMATION

Computer Science and Mathematics Courses: Programming and Data Structures | Linear Algebra | Advanced Calculus | Partial Differential Equation | Probability and Statistics | Transform Calculus | Data Structures and Algorithms
Core and Electives : Economics | Geophysical Signal Processing | Geophysical Field Theory | Introduction to Innovation and Entrepreneurship | Structural Geology | Petrology | Paleontology and Stratigraphy | Methods of Prospecting | Seismology
MOOCs : Machine Learning Specialization by Andrew Ng (Coursera) | Large Language Model Course by Hugging Face | Matlab Onramp by Mathworks | Strivers A2Z Course